Mohammad Hafeez Khan

hafeezmhk6@gmail.com · +91 729 791 9402 · LinkedIn · GitHub · Google Scholar

ABOUT ME

I work on computer vision, mainly on learned representations of 3D scenes. My most recent project is an autoencoder over 3D Gaussian Splatting scenes that compresses a scene into a compact latent and decodes it back, and I built the differentiable rendering pipeline that supervises it in image space, which meant working through camera poses, intrinsics and rendered depth myself. My first degree is in Engineering Physics, and during a summer internship and my bachelor's thesis I characterised materials using X-ray diffraction, UV-visible spectroscopy and other techniques, so imaging physics is not unfamiliar ground. I write my code in PyTorch and train on GPU clusters, and I prefer problems where the method has to work on real measured data.

EDUCATION

Master of Science in Data Science & Artificial Intelligence **Distinction (~85%)**
Queen Mary University of London, England | Sep 2022 – Sep 2023

Bachelor of Technology in Engineering Physics **CGPA: 8.13/10 (Rank 2/20)**
National Institute of Technology Agartala, India | 2016 – 2020

PUBLICATIONS

Peer-Reviewed Articles

- Dotsinski, A., Thakur, U., Ivanov, M., Khan, M. H., & Heuss, M. (2025). On the Generalizability of Competition of Mechanisms: Tracing How Language Models Handle Facts and Counterfactuals. Transactions on Machine Learning Research (TMLR). arXiv:2506.22977.
- Khan, M. H., Thakur, U., Changlani, M., Dotsinski, A., Mahadevan, A. K., Kechagias, I., & Najdenkoska, I. (2025). CLaRE: CLIP with Latent Reconstruction Errors for Generated Face Detection. Proceedings of CCS 2025 (3D-Sec 2025). ACM, Taipei, Taiwan.

RESEARCH PROJECTS

Semantic Scene Completion with 3D Gaussian Splatting **Jan 2026 – Present**
University of Amsterdam, supervised by Yue Li (PhD Student)

- Can3Tok VAE: Perceiver-based autoencoder compressing indoor 3DGS scenes of roughly 40k Gaussians (SceneSplat-7K) into a 512×32 latent and decoding them back into Gaussian primitives, scaled to several thousand unfiltered scenes through a fixed-frame Hilbert ordering, anchor-relative position decoding, and a paired local encoder and token-local decoder. Trained on Snellius H100 GPUs.
- Diagnosed why parameter-space supervision is ill-posed for 3DGS: a fixed decoder slot sees differently oriented surfaces across scenes, so the L2-optimal answer collapses to an isotropic blob, and the (scale, quaternion) parametrisation is gauge-ambiguous, so L2 measures numeric rather than shape distance.
- Now replacing the continuous latent with a discrete VQ-VAE supervised in image space. Built the depth-masked fine-tuning pipeline end to end: camera pose conversion between OpenCV and renderer conventions, normalised intrinsics for a wide-angle capture rig, metric depth rendered through gsplat, and a scene loader for the raw attribute format. Fine-tuning improved held-out reconstruction by 2.2 dB PSNR (17.0 to 19.2) with LPIPS from 0.386 to 0.327.

BLIP3-o DiT: Cross-Modal CLIP Embedding Reproduction **Jul – Aug 2025**
University of Amsterdam Supervised by Dr. Andrii Zadaianchuk

- Built a Diffusion Transformer replicating the BLIP3-o pipeline for reproducing CLIP embeddings from EVA-CLIP features, implementing 3D RoPE, Heun's solver, and grouped-query attention; introduced an EVA adapter for embedding space alignment and trained with FSDP on Snellius H100 GPUs.
- Achieved 89.89% CLIP similarity on training data and 88.76% on MS-COCO validation using sandwich normalisation, layer scaling, and rectified flow matching.

Multi-Modal Semantic Item Identifiers for Generative Recommendation May – Dec 2025
University of Amsterdam Supervised by Kidist Amde Mekonnen (PhD Student)

- Studied multi-modal semantic item identifiers for generative recommendation: fused CLIP visual features with Sentence-T5 text embeddings before RQ-VAE quantisation while holding the TIGER retriever fixed, so that any change in performance is attributable to the visual semantics rather than to architectural changes. Evaluated across eight Amazon categories from the 2014 and 2023 corpora.
- Found the effect of visual signal to be domain-dependent: substantial gains where appearance distinguishes items (Sports & Outdoors, Toys & Games, Video Games) and small, inconsistent or negative effects in text-centric categories such as Software.
- Introduced beyond-accuracy evaluation into the semantic-ID setting (item and brand coverage, Gini exposure inequality, user-level disparity) and identified a trade-off in which broader catalogue coverage can come with more concentrated exposure.

Multilingual Neural Machine Translation for Low-Resource Languages **May – Aug 2023**
Queen Mary University of London, supervised by Prof. Shalom Lappin

- Built a hierarchical transfer learning framework for Indian languages (Bodo, Dogri, Maithili, Konkani) using fine-tuned mBART-50. Achieved BLEU scores of 29–40 through multi-stage training leveraging linguistic similarities.

Nanocomposite Materials for Enhanced Electrical Properties **Jan – May 2020**
National Institute of Technology Agartala, supervised by Prof. Biswajit Saha

- Synthesised PANI/CdS nanocomposites via in-situ polymerisation achieving 13× conductivity improvement. Characterised structural and optical properties using XRD and UV-visible spectroscopy.

TECHNICAL SKILLS

- Programming & Tools: Python, PyTorch, gsplat and differentiable rasterisers, Hugging Face, Transformers, NumPy, scikit-learn, FSDP (distributed training), HPC GPU clusters (H100, SLURM), Git.
- 3D Scene Representations: 3D Gaussian Splatting (training, autoencoding, fine-tuning, evaluation), NeRF and point-based representations, differentiable rendering and image-space losses, depth rendering, camera intrinsics, extrinsics and pose conventions, multi-view geometry (epipolar geometry, homographies, structure from motion, multi-view stereo).
- Deep Learning for Imaging: Encoder-decoder and transformer architectures for reconstruction, reconstruction objectives and their failure modes, image-quality evaluation (PSNR, LPIPS), detection and recognition by adapting pretrained vision encoders.
- Generative & Latent Models: VAEs, VQ-VAEs and residual quantisation, latent diffusion, flow matching and rectified flow, diffusion transformers, autoregressive transformers over discrete tokens.
- Physics & Mathematics: Engineering Physics background (electromagnetism, optics, wave phenomena), X-ray diffraction and UV-visible spectroscopy for materials characterisation, probability and statistics, linear algebra and matrix calculus, Fourier analysis and signal processing, information theory.
- Languages: English (Fluent), Hindi (Native).

RESEARCH INTERESTS

- Learned Representations of 3D Scenes: Storing a scene in a form a network can encode and decode, and working out which parts of the parametrisation actually carry the signal and which are redundant.
- Differentiable Forward Models: Supervising a 3D representation through the imaging process itself rather than on its raw parameters, which is what I built for rendering and what a projection operator does in tomography.
- Transferring Vision Methods Across Imaging Regimes: What breaks when a method designed for reflected light is applied where the measurement is a line integral through the object, since occlusion, opacity and view-dependent appearance stop meaning the same thing.
- Methods That Survive Real Measurements: Validating on measured rather than simulated data, where calibration error, noise and artefacts decide whether a method is usable.